

SM2PNet: A Deep Learning Approach Towards Indian Stock Market Movement Prediction

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Abstract—The objective of this study is to forecast the future price movements of equities listed on the National Stock Exchange (NSE) of India. To anticipate the future price movement, all prior statistical and machine learning techniques have focused on identifying latent temporal dependencies. In actuality, stock prices reflect all available information and are influenced by several external variables, making forecasting difficult. One such major element is the interdependence that exists between stock pairings from the same industry. By incorporating this relational dependence combined with the temporal dependency, we may attain improved prediction performance. In order to demonstrate this, we have created a model, SM2PNet (Stock Market Movement Prediction Network), which takes into consideration the characteristics of highly correlated firms in order to examine their impact on the stock price of the selected target company. We've analysed predefined sector-industry linkages in order to identify the most relevant peer firms. Sequential model long short-term memory (LSTM) is used to represent temporal dependence. The investigation confirms that SM2PNet beats all other available deep learning methods.

Index Terms—Stock market prediction, Correlation analysis, Deep learning, Temporal dependency.

I. INTRODUCTION

Stock market investing provides long- and short-term advantages, passive income, and inflation protection, making it a better investment than fixed deposits, mutual funds, real estate, and commodities. Initial investment is low. Despite huge growth, just 55% of Americans, 33% of Brits, 13% of Chinese, and 3% of Indians trade stocks¹. As of 2021, India, with 138 crore inhabitants, has just 1.2 crore active retail investors, according to the NSE. Risk and uncertainty drive market involvement. Predicting market moves properly reduces risk and boosts rewards. This prompted business and academic academics to construct an efficient prediction model, creating a new interesting inter-disciplinary subject dubbed FinTech (financial technology).

There are two well-known analytical techniques for the stock market, i.e., fundamental [1] and technical [2] analysis. The fundamental analysis approach tries to find the intrinsic value of a commodity or stock by analysing business accounting information such as balance sheets, cash flow statements, profit, and revenue to decide whether a stock is overvalued or undervalued. Whereas the technical analysis approach relies

on historical data (closing price and volume) to build various technical indicators to predict future trends. The selection of technical indicators introduces a subjective bias. According to the Efficient Market Hypothesis (EMH) [3], stock prices reflect all available information and follow a random walk pattern, making the stock market unpredictable. The direct implication is that no analytical approach can beat the market, but certain theories disagree with it. With the advent of artificial intelligence and computation capabilities, it is now possible to model the complex and non-linear nature of stock data.

Methods for stock market forecasting may be broadly grouped into two categories: statistical and machine learning. These statistical methods represent stock market forecasting as a stochastic linear process. In contrast, stock data reveals extremely nonlinear and dynamic patterns. Self-adaptive, data-driven, and non-linear machine learning algorithms have been suggested [4] to eliminate this problem. Due to their ability to model high-dimensional data and generalization capabilities, they do outperform statistical approaches [5].

These methods treat the stock market uniquely and ignore peer and competitor enterprises. Stocks in the same sector have similar trend patterns, thus a change in one company's stock price affects the associated firm. This relational dependency may help examine how external factors affect prediction. One such example is depicted in Figure 1, where an analysis is done during the COVID-19 period. Figure 1(a) shows a positive correlation between the stocks belonging to the same sector, i.e., the IT sector. Whereas Figure 1(b) depicts negative correlation between stocks of healthcare and hotel sector, as during lockdown situation healthcare sector was in demand, but tourism and other sectors were badly affected.

To study the impact of correlation, we have proposed SM2PNet, a deep learning model that consists of two modules, i.e. (1) Correlation analysis module and (2) Temporal Modeling Module. The first module finds the relevant companies affecting the stock price of the target firm using statistical measure. The second module exploits the dependency of historical data points on the present value and predicts the future trend.

The contribution of this work is as follows:

- A proof of the effect of intra-sector correlation between companies on prediction performance.

¹<https://swastika.co.in/blog/population-participating-in-stock-markets-by-country/>

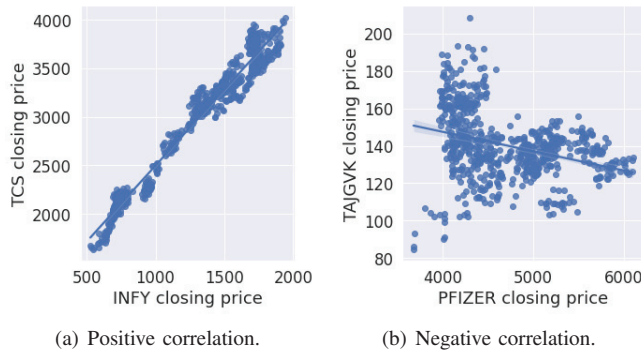


Fig. 1. Correlation analysis between different stock pairs.

- We have proposed SM2PNet, a deep learning model to predict the future price movement of Indian stocks.
- An intra-sectors analysis is performed to verify the effectiveness of SM2PNet.

This paper is organized as follows: Section II reviews the existing literature. Section I describes the dataset. Section IV provides the proof of relation dependency. In Section V SM2PNet is explained in details. Section VI highlights the experimentation details. The result and comprehensive analysis are presented in Section VII followed by the conclusion in Section VIII.

II. RELATED WORK

Traditional statistical approaches include linear regression [6], ARIMA [7], [8], and volatility based GARCH [9] models. These approaches are parametric in nature and suitable only for short term forecasting. Moreover, the requirement of stationary data limits their use in practical life.

To capture the non-linearity, Tripathy [10] proposed Support Vector Machine (SVM) to predict the movement of the closing price after the financial crisis of 2008 for S&P BSE index. Sedighi et al. [11] proposed a hybrid ABC-ANFIS-SVM model to optimize various technical indicators and study the long term fluctuations efficiently. Latha et al. [12] proposed K-nearest Neighbor algorithm to predict the next day's stock price movement and achieved satisfactory results. However, the curse of dimensionality and parameter sensitivity affect the predictive performance of SVM and KNN models.

Inspired by the human brain, deep neural networks have the ability to extract hidden features from the data on their own, which gives them an edge over statistical and machine learning approaches. Farahani et al. [13] proposed a hybrid ANN and meta heuristic approach to predict the stock price of various indices, S&P500, DAX, FTSE100, Nasdaq and DJI. To select the best and most related technical indicators, a heuristics based Genetic Algorithm (GA) was used.

There have also been attempts to utilize state-of-the-art vision models such as convolutional Neural Networks (CNNs) for stock market forecasting. Hiransha et al. [14] applied the CNN-1D model on five different stocks from different sectors of the National Stock Exchange of India and the NYSE

market to predict the next day's closing price. Hoseinzade and Haratizadeh [15] utilized correlations existing among different markets to extract more meaningful relationships and proposed 2D-CNNpred and 3D-CNNpred model. In addition to the historical data, 82 different indicators were used. Mehtab and Sen [16] proposed a multi-channel CNN-based multivariate forecasting model with a five-day horizon to predict the movement of the NIFTY index. It outperformed KNN, SVM and ANN classifiers. CNN models are effective at learning local patterns when the filter size is chosen correctly.

Due to their ability to deal with sequential data, Recurrent Neural Networks (RNNs) are widely used for stock market forecasting. They can exploit the temporal dependency using memory cells. Zhu [17] proposed an RNN model to predict the next day's closing price of Apple stock data. Because RNNs are prone to vanishing and exploding gradients, the error rate rises as the lag value increases. To alleviate this, Long Term Short Memory (LSTM) is proposed, which captures the long range dependency using its powerful gate mechanism. Moghar and Hamiche [18] used an LSTM model to forecast the closing price of two New York Stock Exchange (NYSE) stocks. Wen et al. [19] proposed a PCA-LSTM model to study the effect of technical indicators on the future stock price. Moreover, PCA was utilized to extract the meaningful technical indicators affecting the stock price and reduce the dimensionality. Bhandari et al. [20] proposed multi layer LSTM model to make prediction on the S&P500 index. They have considered macroeconomic, fundamental and technical data. Lin et al. [21] developed an investment strategy based on the prediction results obtained from the GRU model and found that it is more efficient compared to the LSTM model as it requires fewer training parameters to train a model.

Another direction for research is to perform sentiment analysis to predict the stock market. To process such high dimensional unstructured textual data, Natural Language Processing (NLP) based approaches are used. These approaches focus on extracting meaningful semantic information from publicly available sources such as Twitter, news articles, blogs, etc. Gondaliya et al. [22] utilized textual data collected from various online sources to predict the Indian stock market during the COVID-19 pandemic. They have applied three different NLP approaches to extract meaningful features from the textual data and fed them into the classification models. Ko et al. [23] used the BERT model to classify sentiments from news articles and PTT forums for selected Taiwan stocks. To model the temporal dependency, historical data along with the sentiment scores were given to the LSTM model. For sentiment-based approaches, the authenticity of the data remains questionable.

From the literature review, we found that most of the approaches considered only technical indicators or textual data for the prediction and neglected relational features of peer companies. Our aim is to model these features efficiently and study their effect on the prediction results.

III. DATASET DESCRIPTION

In this section, we define the considered stocks from different sectors i.e. Financial Services, Information Technology, Fast Moving Consumer Goods, and Pharmaceutical that cover over 60% of the NIFTY-50 index. We used Python's yfinance² library to collect historical data for these companies from 04-11-2010 to 23-08-2022 (2909 trading days).

We have considered closing price as a feature. 80% of the data is used to train a model, with the remaining 20% used for testing. We have considered peer companies that belong to the same sector as the target firm and are included in the NIFTY-50 index. Table I summarizes the dataset considered for the experimentation.

IV. PROOF OF RELATIONAL DEPENDENCY

In this section, we theoretically prove the dependency of the target firm's price on the stocks belonging to the same sector. In order to prove the relational dependence, we built a simple regression model and evaluated the parameter R^2 to verify the results. It measures the proportion of variance in dependent variable (Target company's closing price) that is explained by the independent variables (Peer companies closing price). The greater value of R^2 is desired to verify goodness of fit. The model is built as:

$$y_{\text{target-company}}^{\text{Close}} = \alpha_1 x_{s_1}^{\text{Close}} + \alpha_2 x_{s_2}^{\text{Close}} + \dots + \alpha_N x_{s_N}^{\text{Close}} \quad (1)$$

Where N is the number of companies belonging to the same sector as the target company. $x_{s_i}^{\text{Close}}$ represents the closing price of i^{th} peer company and α is a learnable parameter. Table II represents the R^2 values obtained from the experiment done.

From the table, we observe that the R^2 values are higher for all the companies, establishing the dependence between the target company and companies belonging to the same sector.

V. FRAMEWORK

In this section, we first formulate the stock market movement prediction problem, and then define SM2PNet.

A. Problem Formulation

Given a series of historical prices of stock S at time instance t denoted as : $X_t = \{x_1, x_2, \dots, x_{t-1}, x_t\}$, where $x_t \in R^D$, our goal is to predict the future price movement y_{t+1} . D represents the dimension of input features. In our case, $D = 1$, as we have considered only closing price as an input feature. This univariate, single-step movement forecasting problem is formulated as follows:

$$y_{t+1} = F(X_t) \quad (2)$$

Where, y_{t+1} is 1 and -1 for upward and downward movement in the stock prices respectively and $F()$ is a user defined prediction function.

²<https://pypi.org/project/yfinance/>

B. Proposed Methodology

An overview of the proposed SM2PNet is depicted in Figure 2. In subsequent sections, each module is discussed in detail.

1) *Correlation analysis module*: This module aims to find the correlation between two companies belonging to the same sector. To do so, we have considered a statistical measure, i.e., Pearson's correlation coefficient, denoted as ρ . It ranges from -1 to 1, where 1 shows a positive correlation, -1 a negative correlation, and 0 indicates no correlation. Given the historical price series of two stock pairs $< i, j >$ for t observation points denoted as $X_i(t) = \{x_i(1), \dots, x_i(t-1), x_i(t)\}$ where $x_i(t)$ denotes the closing price of stock i at time instance t . The correlation value ρ is calculated as follows:

$$\rho_{< i, j >} = \frac{\sum_{k=1}^t [x_i(k) - \hat{X}_i][x_j(k) - \hat{X}_j]}{\sqrt{(\text{var}[X_i])(\text{var}[X_j])}} \quad (3)$$

Where $\hat{X}_i$ and $\text{var}[X_i]$ represents the mean and variance of the series, respectively. The top- K companies having higher correlation value with the target company are selected and their features are concatenated with the historical data of considered firm i to update its embedding as $X'_i(t) = \{x'_i(1), \dots, x'_i(t-1), x'_i(t)\}$ where $x'_i(t) \in R^{K+1}$. These updated embeddings are given to the next module to study the temporal dependency. The results obtained are highlighted in the Section VII.

2) *Temporal modeling module*: This module studies the effect of l previous price values on the current price. To do so, the historical series at time instance t is represented as: $X'_i(t) = \{x'_i(t-l+1), \dots, x'_i(t-1), x'_i(t)\}$. We have considered LSTM model due to its ability to outperform other models on the sequential tasks using powerful gate mechanism. The input is the historical sequence at time instance t denoted as $X'_i(t)$, and the output is predicted by the final layer of the model. The hidden state is updated as:

$$f_t = \sigma(W_f x'_i(t) + V_f h_{t-1} + b_f) \quad (4)$$

$$i_t = \sigma(W_i x'_i(t) + V_i h_{t-1} + b_i) \quad (5)$$

$$o_t = \sigma(W_o x'_i(t) + V_o h_{t-1} + b_o) \quad (6)$$

$$\hat{c}_t = \tanh(W_c x'_i(t) + V_c h_{t-1} + b_c) \quad (7)$$

$$c_t = f_t * ce_{t-1} + i_t * \hat{c}_t \quad (8)$$

$$h_t = o_t * \tanh(c_t) \quad (9)$$

Where, $x'_i(t) \in R^{K+1}$ is an input vector with $K + 1$ dimensional features at time instance t . The forget gate, input gate, output gate, cell state, and hidden state are represented by the variables f_t, i_t, o_t, c_t, h_t . Learnable bias vectors and matrices are represented with the symbols V, W , and b ; * represents element-wise multiplication. The final prediction is obtained as follows:

$$\hat{y}_t = \sigma(W_y h_t + b_y) \quad (10)$$

Where W_y and b_y are learnable parameters. $\hat{y}_t$ denotes the future price movement, 1 for rise in the price and -1 for fall in the price. Figure 3 depicts the LSTM architecture.

TABLE I
DESCRIPTION OF DATASET.

Sector	Sector Weight (%)	Target firm	Ticker Symbol	Peer companies
Financial Services	36.81	HDFC Bank	HDFCBANK	Axis Bank, ICICI Bank, Kotak Bank, SBI, IndusInd Bank
Information Technology	14.7	Tata Consultancy Services	TCS	Infosys, HCL Technologies, Wipro, Tech Mahindra
Fast Moving Consumer Goods	9.02	Hindustan Unilever Limited	HINDUNILVR	ITC, Nestle India, Britannia, Tata Consumer
Pharmaceutical	3.91	Chemical Industrial & Pharmaceutical Laboratories Ltd	CIPLA	Sun Pharmaceutical, Divi's Laboratories, Dr. Reddy's Laboratories

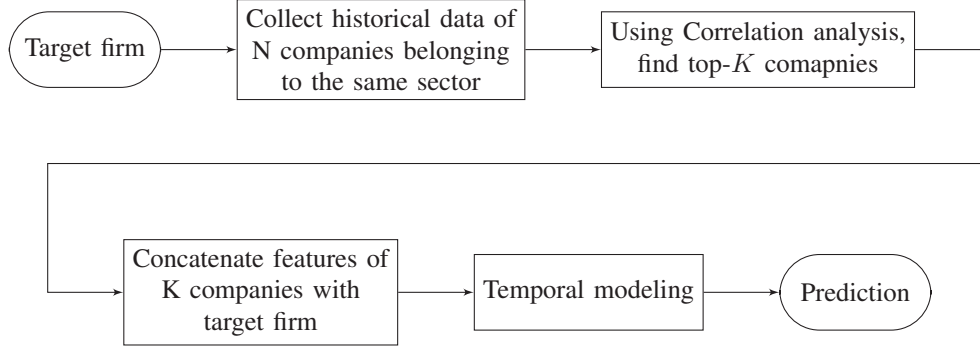


Fig. 2. Proposed SM2PNet framework.

TABLE II
ANALYSIS ON RELATIONAL DEPENDENCY.

Company	R^2 Value
HDFC	0.8412
TCS	0.9167
Hindustan Unilever	0.7806
Cipla	0.8501

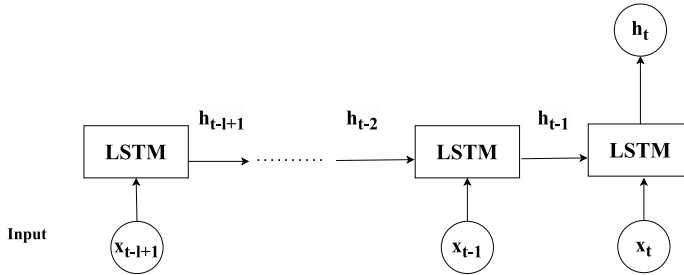


Fig. 3. LSTM Architecture.

VI. EXPERIMENT DETAILS

We considered the following baseline deep learning models to demonstrate the effectiveness of SM2PNet.

A. Baseline models

- **ANN:** It consists of two dense layers with neurons 30 and 32 respectively, followed by final prediction layer.
- **RNN:** It is made up of RNN layer with 128 neurons, followed by two dense layers.

- **CNN:** This model consists of two 1D convolution layers followed by pooling and dense layers. Each conv1D layer consists of 64 filters with the kernel size of 3.
- **LSTM:** It consists of two stacked LSTM layers followed by a dense layer.
- **SM2PNet:** It consists of two modules: a correlation module and an LSTM model with two layers and a final prediction layer.

For uniformity, all models have a considered lag value of 30, and the Adam optimizer is used to train models.

B. Evaluation Parameters

As a stock movement prediction task is considered a classification problem, we have considered accuracy, precision, and recall to evaluate the models. These are calculated using the confusion matrix shown in Table III.

TABLE III
CONFUSION MATRIX.

		Actual values		Total
		Positive	Negative	
Predicted values	Positive	TP	FP	TP + FP
	Negative	FN	TN	FN + TN
Total		TP + FN	FP + TN	N

- **Accuracy:** Accuracy (Equation 11) measures the fraction of predictions classified correctly by the model. It ranges from 0% to 100%.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

- **Precision:** Precision reflects the ability of the model to predict a specific category, and ranges 0 to 1.

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

- **Recall:** Recall identifies how many times the model was able to detect a specific category. It ranges 0 to 1.

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

- **F-score:** It balances precision and recall scores and measures overall model performance. It ranges 0 to 1.

$$F\text{-score} = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (14)$$

Higher values of above parameters are preferred.

VII. RESULT AND ANALYSIS

In this section, we represent the experiment results for each sector. We have performed the correlation analysis with IT sector companies.

A. Analysis on IT sector

Table IV represents the correlation analysis of IT sector companies. For our target IT firm, TCS, Infosys, HCL Technologies, and Tech Mahindra are considered as as correlated companies and concatenate their features to update the embedding.

TABLE IV
CORRELATION ANALYSIS ON IT SECTOR.

Company	TCS	INFY	HCLTECH	WIPRO	TECHM
TCS	1.0000	0.9656	0.9777	0.9225	0.9582
INFY	0.9656	1.0000	0.9632	0.9706	0.9408
HCLTECH	0.9777	0.9632	1.0000	0.9387	0.9679
WIPRO	0.9225	0.9706	0.9387	1.0000	0.9336
TECHM	0.9582	0.9408	0.9679	0.9336	1.0000

Figure 4 represents the trend patterns of all IT sector companies.

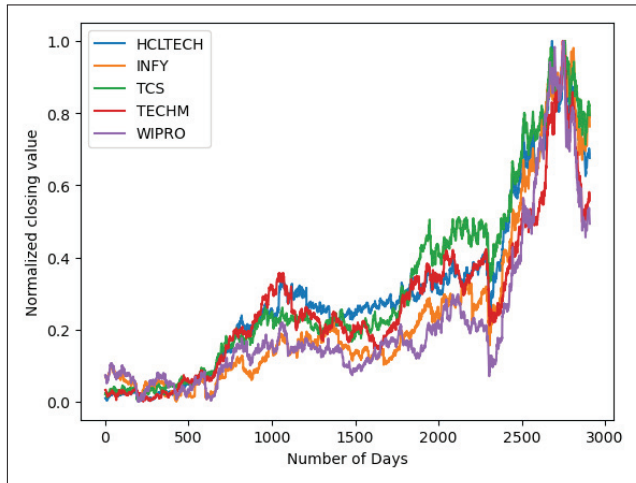


Fig. 4. Correlation analysis on IT sector.

Table V evaluates the performance of various deep learning models for TCS. We can observe that our approach outperforms all other models.

TABLE V
IT SECTOR PREDICTION RESULTS.

Model	Accuracy	Precision	Recall	F-score
ANN	48.91	0.5153	0.5207	0.5180
CNN	49.09	0.5174	0.5138	0.5156
RNN	49.27	0.5179	0.5483	0.5327
LSTM	50.36	0.529	0.5345	0.5317
SM2PNet	52	0.5454	0.5379	0.5416

B. Analysis on Banking sector

Table VI represents the correlation analysis for the banking sector. For our target company, HDFC Bank, we will consider ICICI and Kotak as correlated companies.

TABLE VI
CORRELATION ANALYSIS ON BANKING SECTOR.

Company	HDFCBANK	ICICIBANK	SBIN	KOTAKBANK	AXISBANK	INDUSINDBK
HDFCBANK	1.0000	0.9025	0.7354	0.9928	0.8922	0.6020
ICICIBANK	0.9025	1.0000	0.8889	0.9024	0.8150	0.3372
SBIN	0.7354	0.8889	1.0000	0.7214	0.7600	0.3824
KOTAKBANK	0.9928	0.9024	0.7214	1.0000	0.8941	0.5853
AXISBANK	0.8922	0.8150	0.7600	0.8941	1.0000	0.7269
INDUSINDBK	0.6021	0.3372	0.3825	0.5853	0.7269	1.0000

Figure 5 depicts the trends exhibited by banking sector companies, and we can verify that Kotak and ICICI Bank are closely related to HDFC Bank.

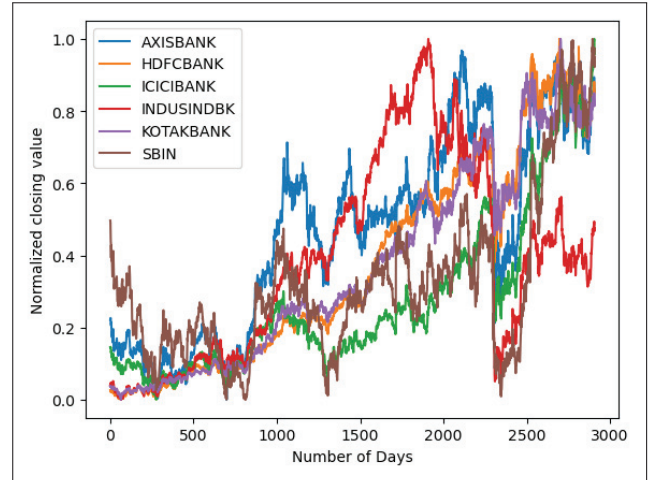


Fig. 5. Correlation analysis on Banking sector.

Table VII represents the prediction results for the considered banking sector company. SM2PNet outperforms other models with higher accuracy.

C. Analysis on FMCG sector

Table VIII represents the correlation analysis for FMCG sector companies. For our considered firm, Hindusan Unilver Ltd., we will take into account Nestle India and Britannia to update the embedding.

TABLE VII
BANKING SECTOR PREDICTION RESULTS.

Model	Accuracy	Precision	Recall	F-score
ANN	46.91	0.4840	0.4806	0.4823
CNN	50.54	0.5197	0.5123	0.5160
RNN	48.90	0.5034	0.5194	0.5113
LSTM	51.45	0.5292	0.5124	0.5207
SM2PNet	52	0.5331	0.5406	0.5368

TABLE VIII
CORRELATION ANALYSIS ON FMCG SECTOR.

Company	HINDUNILVR	ITC	NESTLEIND	BRITANNIA	TATACONSUM
HINDUNILVR	1.0000	0.4210	0.9726	0.9717	0.8633
ITC	0.4210	1.0000	0.2678	0.4924	0.1912
NESTLEIND	0.9726	0.9543	1.0000	0.9171	0.9180
BRITANNIA	0.9717	0.4924	0.9171	1.0000	0.7896
TATACONSUM	0.8633	0.1912	0.9180	0.7896	1.0000

The trend pattern of FMCG sector companies is depicted in Figure 6. The results obtained for FMCG sector companies are represented in the Table IX.

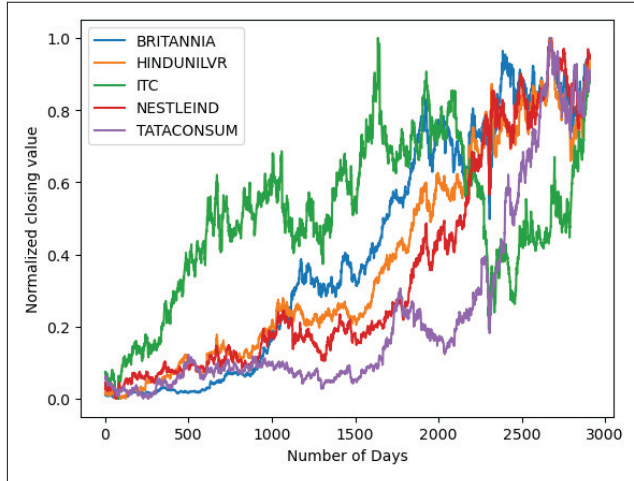


Fig. 6. Correlation analysis on FMCG sector.

TABLE IX
FMCG SECTOR PREDICTION RESULTS.

Model	Accuracy	Precision	Recall	F-score
ANN	45.27	0.4432	0.4481	0.4456
CNN	44.54	0.439	0.4667	0.4524
RNN	47.27	0.465	0.4925	0.4783
LSTM	46.91	0.4618	0.4926	0.4767
SM2PNet	48.72	0.479	0.5074	0.4928

D. Analysis on Pharmaceutical sector

The correlation analysis for the pharmaceutical sectors is represented in Table X. For target company Cipla, we will consider Dr. Reddy's Laboratories and Divi's Laboratories as correlated companies. Figure 7 depicts the trend patterns for pharmaceutical sector companies.

TABLE X
CORRELATION ANALYSIS ON PHARMA SECTOR.

Company	CIPLA	SUNPHARMA	DIVISLAB	DRREDDY
CIPLA	1.0000	0.6950	0.8634	0.8825
SUNPHARMA	0.6950	1.0000	0.3938	0.6521
DIVISLAB	0.8634	0.3938	1.0000	0.8505
DRREDDY	0.8825	0.6521	0.8505	1.0000

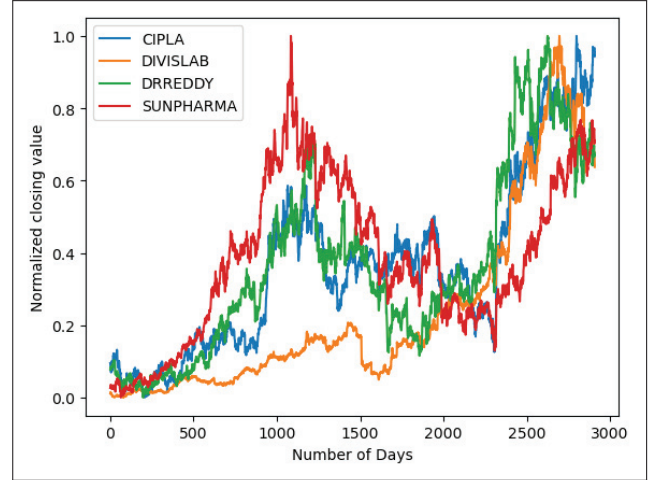


Fig. 7. Correlation analysis on Pharma sector.

Table XI represents the prediction results for pharmaceutical sector companies. SM2PNet fails for the recall and F-score parameters. This can be explained by the fact that the correlation value between pharmaceutical sector is not that much higher. This can be resolved by finding more reliable companies that can identify the underlying data distribution of the target company.

TABLE XI
PHARMACEUTICAL SECTOR RESULTS.

Model	Accuracy	Precision	Recall	F-score
ANN	43.45	0.4437	0.4516	0.4476
CNN	43.091	0.4397	0.4444	0.4420
RNN	46.72	0.4746	0.4695	0.4720
LSTM	45.81	0.4641	0.4408	0.4521
SM2PNet	47.45	0.4812	0.4588	0.4697

From the analysis done, we confirm that the correlation existing between stocks has an impact on prediction performance. SM2PNet is able to learn data distribution of different companies efficiently.

VIII. CONCLUSION

We proposed SM2PNet, a model to forecast the future price movement of Indian equities from the Banking, IT, FMCG, and Pharma sectors. Relational dependency, the target firm's stock price reliance on its peers is examined in this research. All prior techniques have treated a stock as a separate entity, although there is a link between stocks that may explain external events. To determine a target firm's peers, we used the

pre-defined sector connection. We selected the most significant firms using a correlation technique and revised the target firm's feature embedding to decrease noise and instability. LSTM effectively models long-range dependence and temporal dependency and achieves best performance.

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